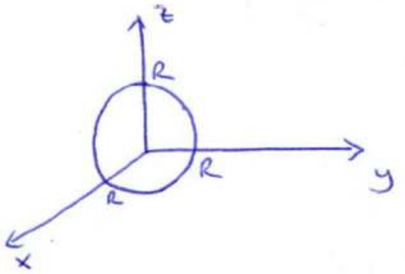
TEMEL YÜZEYLER

Küre: $x^2+y^2+z^2=R^2$ => Yarıçapı R olan, orjin merkezli küre



$(x-a)^2+(y-b)^2+(z-c)^2=R^2$ => Merkezi $M(a,b,c)$ noktası olan R yarıçaplı küre

R bölgesi dikdörtgen olursa

Ortalama Değer

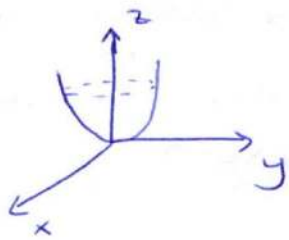
$a \leq x \leq b$

$c \leq y \leq d$

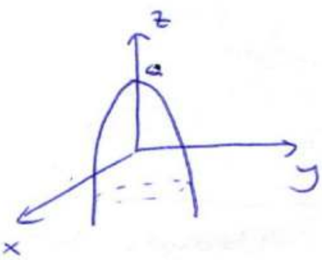
$$\frac{1}{(b-a)(d-c)} \cdot \iint_R f(x,y)\,dA$$

Paraboloid:

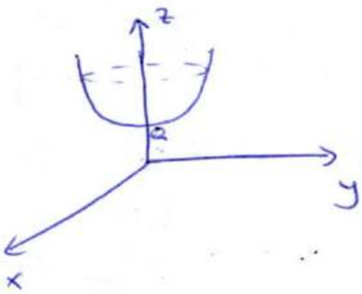
① $z=ax^2+by^2$ $(a,b>0)$ => Orjin tepe noktalı, kolları yukarı paraboloid



② $z=a-x^2-y^2$ => $(0,0,a)$ tepe noktalı, kolları aşağı paraboloid

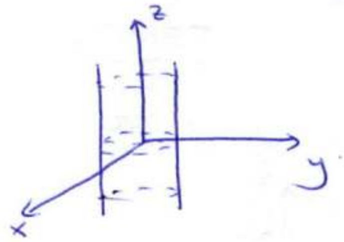


③ $z=a+x^2+y^2$ => $(0,0,a)$ tepe noktalı, kolları yukarı



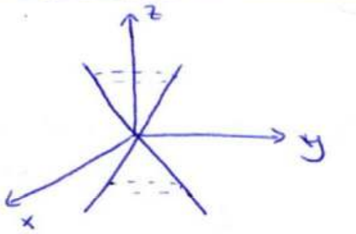
Silindir:

$x^2+y^2=r^2 \Rightarrow z$ boyunca uzanan silindir $\Rightarrow$

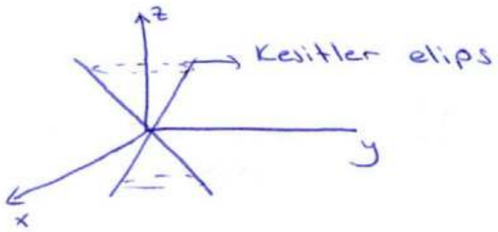


Koni:

* $x^2+y^2=z^2 \Rightarrow$ Dairesel Koni $\Rightarrow$

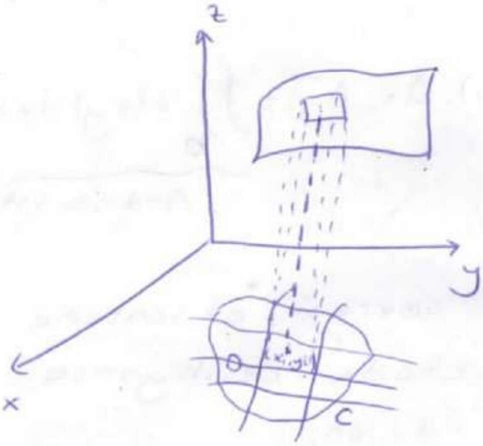


* $\frac{x^2}{a^2}+\frac{y^2}{b^2}=z^2 \Rightarrow$ Eliptik Koni $\Rightarrow$



2 Katlı İntegral

$z=f(x,y)$ fonksiyonu xOy düzleminde C eğrisiyle sınırlı, kapalı bir D bölgesinde tanımlı ve sürekli olsun. D bölgesini, alanları ΔA_i $(i=1,2,\dots,n)$ olan kısmi bölgelere ayırıp, bu bölgelerden keyfi (x_i,y_i) noktaları seçelim.



$D \to D_1,\ D_2,\ \dots,\ D_n$

$D_1 \downarrow \Delta A_1 \downarrow P_1(x_1,y_1) \downarrow z_1=f(x_1,y_1)$

$D_2 \downarrow \Delta A_2 \downarrow P_2(x_2,y_2) \downarrow z_2=f(x_2,y_2)$

$\dots$

$D_n \downarrow \Delta A_n \downarrow P_n(x_n,y_n) \downarrow z=f(x_n,y_n)$

$$f(x_1,y_1).\Delta A_1+f(x_2,y_2).\Delta A_2+\dots+f(x_n,y_n).\Delta A_n=\sum_{i=1}^{n} f(x_i,y_i).\Delta A_i$$

Bu toplam, tabanı ΔA_i ve yüksekliği $f(x_i,y_i)$ olan silindir elemanlarının hacimleri toplamıdır.

ΔA_i alanlarının herbirinin sıfıra yaklaşması halinde bu toplamın limitine $z=f(x,y)$ fonksiyonunun D bölgesinde iki katlı integrali denir ve

$$\lim_{n\to\infty}\sum_{i=1}^{n} f(x_i,y_i)\,\Delta A_i=\iint_D f(x,y)\,dA = V$$ şeklinde gösterilir.

* Bu integralin değeri, D bölgesinin çevresi üzerinde, üstten $z=f(x,y)$ yüzeyi, alttan $z=0$ düzleminin sınırladığı hacime eşit olur.

Bu limit D bölgesinin kısmi bölgelere bölünüş şekline ve P_i noktalarının ΔA_i içindeki seçiliş şekline bağlı değildir.

* Eğer D bölgesi eksenlere paralel doğrularla kısmi bölgelere ayrılırsa, kısmi bölgeler birer dikdörtgen olur ve bu dikdörtgenlerin alanları

$\Delta A_i = \Delta x_i . \Delta y_i$ ve limit de

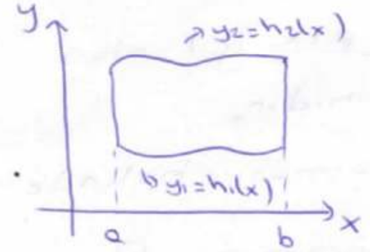
$$\lim_{n\to\infty} \sum_{i=1}^{n} f(x_i,y_i)\Delta A_i = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i,y_i).\Delta x_i \Delta y_i = \underbrace{\iint_D f(x,y)\,dx\,dy}_{\text{Ardışık integral}}$$

olur. D bölgesine integrasyon bölgesi denir.

Düzgün bölge: Eğer D bölgesinin çevresi, eksenlere dik doğrularla en çok 2 noktada kesiliyorsa böyle bölgeye düzgün bölge denir.

Dik Kesitleri Kullanarak 2 Katlı İntegral Hesaplamak

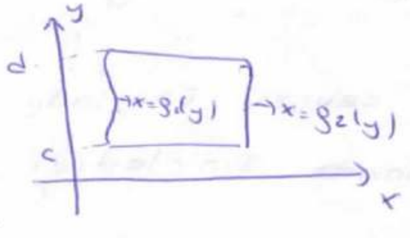
* Bölge x'e göre düzgündür.
* Bölge x'e dik doğrularla taranır.



$$\iint_D f(x,y)\,dA = \int_a^b \int_{h_1(x)}^{h_2(x)} f(x,y)\,dy\,dx$$

Yatay Kesitler ile 2 Katlı İntegral Hesaplamak

* Bölge y'ye göre düzgündür
* Bölge y'ye dik doğrularla taranır.



$$\iint_D f(x,y)\,dA = \int_c^d \int_{g_1(y)}^{g_2(y)} f(x,y)\,dx\,dy$$

(*) $D: \begin{Bmatrix} x=0 \\ x=2 \\ y=0 \\ y=x^2 \end{Bmatrix}$ $I=\iint_D y\,dx\,dy$ integralini

I.3

a) x'e göre düzgün bölge

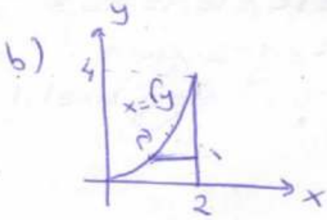
b) y'ye " " " olarak hesaplayın

a)

$$I=\int_0^2\int_0^{x^2} y\,dy\,dx = \int_0^2\left(\frac{y^2}{2}\Big|_0^{x^2}\right)dx$$

$$=\int_0^2 \frac{x^4}{2}\,dx = \frac{x^5}{10}\Big|_0^2 = \frac{16}{5}$$

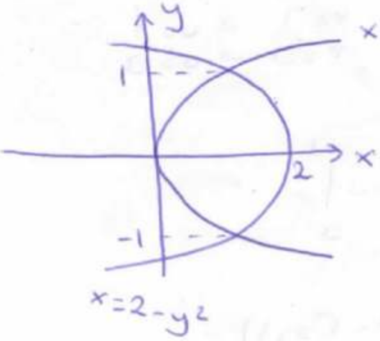
b)



$$I=\int_0^4\int_{\sqrt{y}}^2 y\,dx\,dy = \int_0^4\left(yx\Big|_{\sqrt{y}}^2\right)dy$$

$$=\int_0^4 (2y-y^{3/2})\,dy = y^2-\frac{y^{5/2}}{5/2}\Big|_0^4 = \frac{16}{5}$$

(*) $D: \begin{Bmatrix} x=y^2 \\ x=2-y^2 \end{Bmatrix}$ bölgesinde $f(x,y)=1+5y$ fonksiyonunun integralini hesaplayın.



$\left.\begin{matrix} x=y^2 \\ x=2-y^2 \end{matrix}\right\}$ $y^2=2-y^2 \Rightarrow y=\mp 1$

I. Yol

$$\iint_D (1+5y)\,dA = \int_{-1}^1\int_{y^2}^{2-y^2}(1+5y)\,dx\,dy \rightarrow$$ y'ye göre düzgün bölge aldık

$$=\int_{-1}^1 (x+5xy)\Big|_{y^2}^{2-y}\,dy$$

$$=\int_{-1}^1 (2-2y^2+10y-10y^3)\,dy = \frac{8}{2}$$

$y^2 \le x \le 2-y^2$

$-1 \le y \le 1$

$$\int_{-1}^1\int_{y^2}^{2-y^2} 1+5y\,dx\,dy$$

$$x+5yx\Big|_{y^2}^{2-y^2}$$

$$2-y^2+10y-5y^3-y^2-5y^3$$

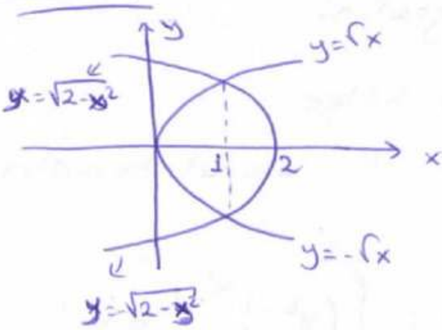
$$\int_{-1}^1 -10y^3-2y^2+10y+2$$

$$-\frac{10y^4}{4}-\frac{2y^3}{3}+5y^2+2y\Big|_{-1}^1$$

$$-\cancel{\frac{5}{2}}-\cancel{\frac{2}{3}}+7+\cancel{\frac{5}{2}}+\cancel{\frac{2}{3}}-5+2$$

$2-3=\frac{4}{7}$

2.4ol



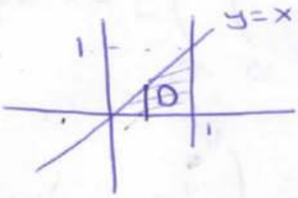
x'e göre düzgün bölge olursak

$$I=\int_0^1\int_{-\sqrt{x}}^{\sqrt{x}}(1+5y)\,dy\,dx+\int_1^2\int_{-\sqrt{2-x^2}}^{\sqrt{2-x^2}}(1+5y)\,dy\,dx$$

İntegrasyon Sırasını Değiştirme

⊗ $I=\int_0^1\int_y^1 \sin x^2\,dx\,dy=?$ $\Rightarrow$ $\int_y^1 \sin x^2\,dx$ hesaplanamaz. İntegrasyon sırası değişmeli!

D: $y=1$ $y=0$
$x=y$ $x=1$



$0\le y\le x$
$0\le x\le 1$

$$I=\int_0^1\int_0^x \sin x^2\,dy\,dx=\int_0^1\left(y\sin x^2\right)\Big|_0^x dx$$

$$\int_0^1\int_0^x \sin x^2\,dy\,dx$$

$$y.\sin x^2\Big|_0^x$$

$$=\int_0^1 (x\sin x^2)\,dx$$

$x^2=u$ $2x\,dx=du$
$x=1\rightarrow u=1$
$x=0\rightarrow u=0$

$$=\int_0^1 \frac{\sin u}{2}\,du=-\frac{1}{2}\cos u\Big|_0^1$$

$$\int_0^1 x.\sin x^2$$

$x^2=u$

$2x\,dx=du$

$$=-\frac{1}{2}\cos 1+\frac{1}{2}=\frac{1}{2}(1-\cos 1)$$

$$\int_0^1 \frac{\sin u}{2}\,du$$

$$\frac{-\cos x^2}{2}\Big|_0^1$$

$$\frac{-\cos 1}{2}+\frac{1}{2}=\frac{1-\cos 1}{2}$$

(15)

(*) $\int_0^1\int_x^1 e^{x/y}\,dy\,dx = ?$ D: $x=1$ $y=1$ $x=0$ $y=x$

$0\le x\le y$

$0\le y\le 1$

$\int_0^1\int_0^y e^{x/y}\,dx\,dy$

$y\cdot e^{\frac{x}{y}}\Big|_0^y$

$I=\int_0^1\int_0^y e^{x/y}\,dx\,dy=\int_0^1 \frac{e^{x/y}}{\frac{1}{y}}\Big|_0^y dy=\int_0^1 (ey-y)\,dy$ $\int_0^1 e.y-y\,dy$

$=\frac{ey^2}{2}-\frac{y^2}{2}\Big|_0^1=\frac{e-1}{2}$

$\frac{ey^2}{2}-\frac{y^2}{2}\Big|_0^1$ $\frac{e-1}{2}$

İki Katlı İntegralle Düzlem Alanların Hesabı

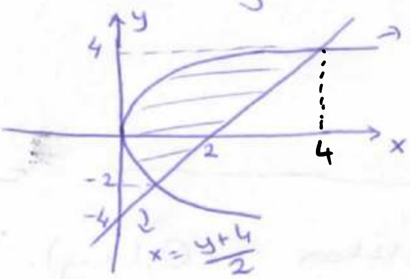
$\iint_D f(x,y)\,dx\,dy$ integralinde $f(x,y)=1$ ise $\iint_D dx\,dy$ integrali D bölgesinin alanını verir.

$$\boxed{A=\iint_D dx\,dy}$$

(*) $y^2=4x$ ile $y=2x-4$ doğrusu arasında kalan alanı 2 katlı integralle hesaplayın.

$\frac{y^2}{4}\le x\le\frac{y+4}{2}$

$-2\le y\le 4$



$A=\int_{-2}^{4}\int_{y^2/2}^{\frac{y+4}{2}} dx\,dy=\int_{-2}^{4}\left(\frac{y+4}{2}-\frac{y^2}{2}\right)dy=9$

$\int_{-2}^{4}\int_{\frac{y^2}{2}}^{\frac{y+4}{2}} dx\,dy$ $\int_{-2}^{4}\frac{y+4-y^2}{2}\,dy$ $15-12=\frac{3}{7}$

$4x=(2x-4)^2$

$4x=4x^2-16x+16$

$x^2-5x+4=0$ $x=4\to y=4$

-4 -1 $x=1\to y=-2$

$\frac{y^2}{4}+2y-\frac{y^3}{6}\Big|_{-2}^{4}$

$4+8-\frac{64}{6}-1+4-\frac{8}{6}$ $\frac{-72}{6}=-12$

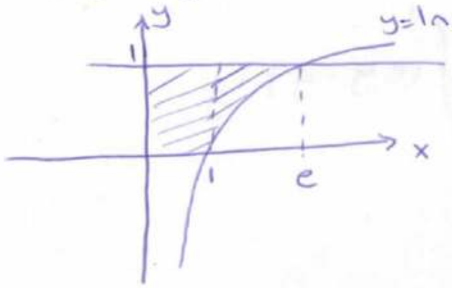
$0 \le x \le e$
$\le y \le$

$\boxed{\text{i.6}}$

(*) $y=\ln x$, $y=1$, $x=0$, $y=0$ eğrilerinin sınırladığı alanı veren 2 katlı integralli

a) x'e göre düzgün bölge olarak

b) y'ye göre " " " yazınız.



a) $A_D = \iint_D dy\,dx$

$$= \int_0^1 \int_0^1 dy\,dx + \int_1^e \int_{\ln x}^1 dy\,dx$$

b)

$$A = \int_0^1 \int_0^{e^y} dx\,dy$$

İki Katlı İntegralde Hacim Hesabı

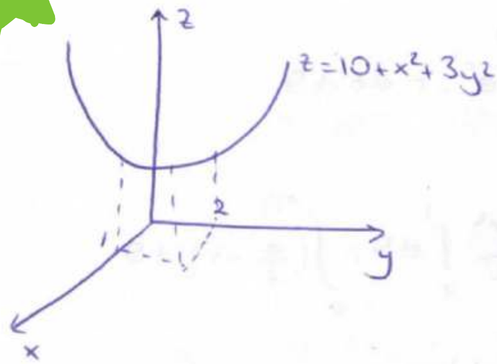
① $f(x,y)$ bir D bölgesi üzerinde pozitif bir fonksiyon olsun. Bu durumda üstten $f(x,y)$, alttan $z=0$ ile sınırlı cismin D bölgesi üzerinde oluşturduğu cismin hacmi

$$V = \iint_D f(x,y)\,dx\,dy \quad \text{dır.}$$

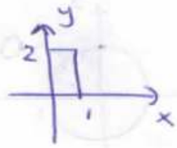
② Eğer cisim üstten $z=Q_2(x,y) \geq 0$, alttan $z=Q_1(x,y)$ ile sınırlı ise, bu yüzeylerin xy düzlemindeki izdüşümü olan D taban olmak üzere, hacim; bu yüzeylerle sınırlı cisimlerin hacimleri farkına eşittir.

$$V = \iint_D (Q_2(x,y) - Q_1(x,y))\,dx\,dy$$

⊛ Üstten $z=10+x^2+3y^2$ paraboloidi ve alttan $z=0$ da $R:\left\{\begin{matrix}0\le x\le 1\\ 0\le y\le 2\end{matrix}\right\}$ dikdörtgeni ile sınırlı bölgenin hacmi?



$$V=\iint_R (10+x^2+3y^2)\,dy\,dx$$



$$=\int_0^1\int_0^2 (10+x^2+3y^2)\,dy\,dx$$

$$=\int_0^1 (10y+x^2y+y^3)\Big|_0^2\,dy$$

$$=\int_0^1 (20+2x^2+8)\,dx = 28x+\frac{2x^3}{3}\Big|_0^1=\frac{86}{3}$$

İki Katlı İntegrallerin Kutupsal Koordinatlara Dönüştürülerek Hesabı

$\iint_D f(x,y)\,dx\,dy$ iki katlı integralini $\left\{\begin{matrix}x=r\cos\theta\\ y=r\sin\theta\end{matrix}\right\}$ dönüşümü yaparak hesaplarsak; bu durumda D bölgesi, $r=f_1(\theta)$, $r=f_2(\theta)$ eğrileri ve $\theta=\alpha$, $\theta=\beta$ doğrularının sınırladığı bölge olsun. Bu dönüşümle $dx\,dy=r\,dr\,d\theta$ ve

$$\iint_D f(x,y)\,dx\,dy=\int_\alpha^\beta\int_{f_1(\theta)}^{f_2(\theta)} f(r\cos\theta, r\sin\theta).\,r\,dr\,d\theta \quad \text{olur.}$$

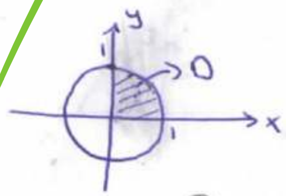
★ Kutupsal dönüşümle D bölgesinin alanı ise:

$$\iint_D dx\,dy=\int_\alpha^\beta\int_{f_1(\theta)}^{f_2(\theta)} r\,dr\,d\theta \quad \text{olur.}$$

18

(*) $I = \int_0^1 \int_0^{\sqrt{1-x^2}} e^{x^2+y^2}\, dy\, dx = ?$

$D: \; x=1 \quad x=0$
$y=\sqrt{1-x^2} \quad y=0$



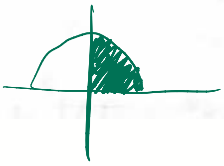
$x = r\cos\theta$
$y = r\sin\theta$
$x^2+y^2 = r^2$
$dx\,dy = r\,dr\,d\theta$

$$I = \int_0^{\pi/2}\int_0^1 e^{r^2}\cdot r\,dr\,d\theta$$

$y = \sqrt{1-x^2}$

$$= \int_0^{\pi/2} \frac{e^{r^2}}{2}\Big|_0^1 d\theta = \int_0^{\pi/2}\left(\frac{e}{2}-1\right)d\theta$$

$$= \frac{\pi}{4}(e-1)$$



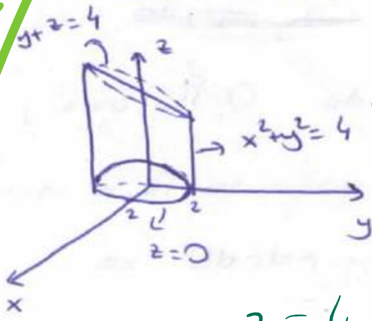
$0 \le r \le 1$
$0 \le \theta \le \frac{\pi}{2}$

$$\int_0^{\pi/2}\int_0^1 e^{r^2}\cdot r\,dr\,d\theta \qquad \frac{e^{r^2}}{2}\Big|_0^1$$

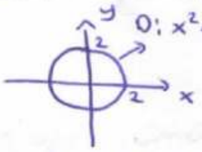
$$\frac{e}{2}-\frac{1}{2}\Big|_0^{\pi/2}$$

$$\left(\frac{e-1}{4}\right)\cdot\pi$$

(*) $x^2+y^2=4$ silindiri ve $y+z=4$, $z=0$ düzlemleri tarafından sınırlanan cismin hacmini hesaplayınız.



$D: x^2+y^2=4$



$$V = \iint_D (4-y)\,dy\,dx$$

$0 \le r \le 2$
$0 \le \theta \le 2\pi$

$x = r\cos\theta$
$y = r\sin\theta$
$x^2+y^2=r^2$
$dx\,dy = r\,dr\,d\theta$

$$= \int_0^{2\pi}\int_0^2 (4-r\sin\theta)\,r\,dr\,d\theta$$

$z = 4-y$

$y = r\sin\theta$

$$\int_0^{2\pi}\int_0^2 (4-y)\,dx\,dy$$

$$= \int_0^{2\pi}\left(2r^2-\frac{r^3}{3}\sin\theta\Big|_0^2\right)d\theta = \int_0^{2\pi}\left(8-\frac{8}{3}\sin\theta\right)d\theta$$

$$= 8\theta + \frac{8}{3}\cos\theta\Big|_0^{2\pi}$$

$$= 16\pi$$

$(4-r\sin\theta)r$

$$\int_0^{2\pi}\int_0^2 4r - r^2\sin\theta\,dr\,d\theta$$

$$2r^2-\frac{r^3}{3}\sin\theta\Big|_0^2$$

$$8-\frac{8}{3}\sin\theta$$

$$16\pi+\frac{8}{3}-\frac{8}{3} = 16\pi$$

2π

$$8\theta+\frac{8}{3}\cos\theta\Big|_0$$

$$\iint_{x^2+(y-2)^2=4} 4-x\,dx\,dy = \int_0^{\pi}\int_0^{4\sin\theta}(4-r\cos\theta)r\,dr\,d\theta$$

19

(*) $x^2+y^2-4y=0$ silindiri, $x+z=4$, $z=0$ düzlemleri arasındaki cismin hacmini veren 2 katlı integral?

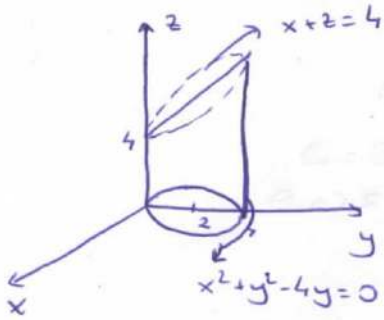
$r^2=4r\sin\theta$

$x^2+y^2-4y+4-4=0 \Rightarrow x^2+(y-2)^2=4$

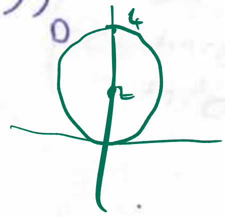
$0\le r\le 4\sin\theta$

$0\le\theta\le\pi$

D: $x^2+(y-2)^2=4$



$$V=\iint_D (4-x)\,dy\,dx$$



$$\begin{pmatrix} x=r\cos\theta \\ y=r\sin\theta \\ dx\,dy=r\,dr\,d\theta \\ x^2+y^2=r^2 \end{pmatrix}$$

$x^2+y^2-4y=0$

$r^2-4r\sin\theta=0$

$r=0 \quad r=4\sin\theta$

$$V=\iint_D (4-x)\,dy\,dx=\int_0^{\pi}\int_0^{4\sin\theta}(4-r\cos\theta)r\,dr\,d\theta$$

(*) $x^2+y^2+z^2=4$ ün altında $3z=x^2+y^2$ nin üstünde kalan cismin hacmi?

$$\iint \sqrt{4-x^2-y^2}-\frac{x^2+y^2}{3}\,dx\,dy$$

$z^2+3z-4=0$

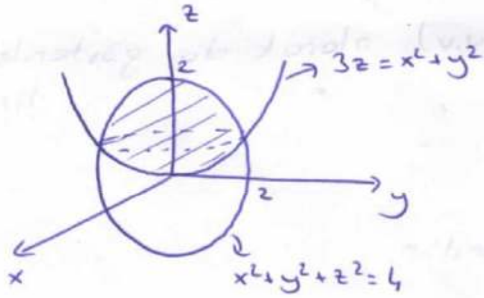
$z=-4$

$z=1$

$x^2+y^2=3$

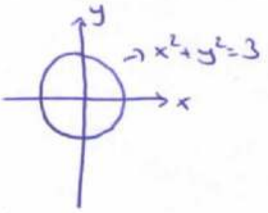
$x^2+y^2+z^2=4 \to$ Küre

$3z=x^2+y^2 \to$ Paraboloid



$$V=\iint_D\left(\sqrt{4-x^2+y^2}-\frac{x^2+y^2}{3}\right)dy\,dx$$

$\left.\begin{matrix}3z=x^2+y^2 \\ x^2+y^2+z^2=4\end{matrix}\right]$ $z^2+3z-4=0$ $\boxed{z=1} \to x^2+y^2=3$ izdüşüm bölgesi D



$$\begin{pmatrix} x=r\cos\theta \\ y=r\sin\theta \\ dx\,dy=r\,dr\,d\theta \\ x^2+y^2=r^2 \end{pmatrix} \Rightarrow V=\int_0^{2\pi}\int_0^{\sqrt3}\left(\sqrt{4-r^2}-\frac{r^2}{3}\right)r\,dr\,d\theta$$

$0\le r\le\sqrt3 \quad 0\le\theta\le 2\pi$

$$=\frac{19}{6}\pi$$

$$\int_0^{2\pi}\int_0^{\sqrt3}\left(\sqrt{4-r^2}-\frac{r^2}{3}\right)r\,dr\,d\theta=\frac{19}{6}\pi$$

110

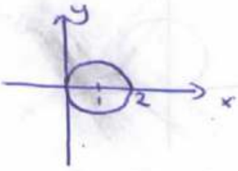
(*) $\iint_D (x^2+y^2)\,dx\,dy$ integralini $D: x^2+y^2=2x$ bölgesinde kutupsal koordinatlara dönüştürerek yazınız.

$(x-1)^2+y^2=1$

$r^{\cancel{2}}=2\cancel{r}\cos\theta$

$r=2\cos\theta$

$x^2+y^2-2x=0 \quad \rightarrow (x-1)^2+y^2=1$

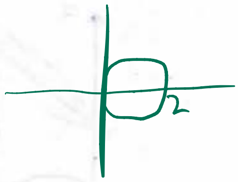


$$\left.\begin{aligned} x&=r\cos\theta\\ y&=r\sin\theta\\ dx\,dy&=r\,dr\,d\theta\\ x^2+y^2&=r^2 \end{aligned}\right\}$$

$x^2+y^2-2x=0$

$r^2-2r\cos\theta=0$

$r=0 \qquad r=2\cos\theta$



$0\le r\le 2\cos\theta$

$-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$

$$I=\int_{-\pi/2}^{\pi/2}\int_0^{2\cos\theta} r^2\cdot r\,dr\,d\theta$$

$$\boxed{\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_0^{2\cos\theta} r^3\,dr\,d\theta}$$

- **Jakobien Determinantı**

$x=g(u,v)$ ve $y=h(u,v)$ koordinat dönüşümünün Jakobien determinantı veya Jakobieni şöyledir:

$$J(u,v)=\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}$$

$J(u,v)$, $\dfrac{\partial(x,y)}{\partial(u,v)}$ olarak da gösterilir.

NOT: $\dfrac{\partial(x,y)}{\partial(u,v)}=\dfrac{1}{\dfrac{\partial(u,v)}{\partial(x,y)}}$ dir.

$$=\frac{1}{\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}}$$

İki Katlı İntegralde Değişken Dönüşümü

(İ.K.11)

$\iint_D f(x,y)\,dx\,dy$ integralinde $x=h(u,v)$, $y=g(u,v)$ değişken dönüşümü yapılırsa D bölgesi bir D' bölgesine dönüşür. Bu halde,

$$\iint_D f(x,y)\,dx\,dy = \iint_{D'} f(h(u,v), g(u,v)).\,|J(u,v)|.\,du\,dv \quad \text{olur.}$$

$$\left.\begin{array}{l} x=h(u,v) \\ y=g(u,v) \end{array}\right\} \Rightarrow \boxed{\begin{array}{l} D \longrightarrow D' \\ dx\,dy \longrightarrow |J(u,v)|\,du\,dv \end{array}}$$

(*) $\left\{\begin{array}{l} x=r\cos\theta \\ y=r\sin\theta \end{array}\right\}$ kutupsal dönüşümü ile $dx\,dy = r\,dr\,d\theta$ olduğunu gösteriniz.

$$J(r,\theta) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = \underline{\underline{r}}$$

$$dx\,dy = |J(r,\theta)|\,dr\,d\theta = \underline{\underline{r\,dr\,d\theta}}$$

(*) $D: \left\{\begin{array}{l} x-y=1 \\ x-y=3 \\ xy=1 \\ xy=4 \end{array}\right\}$ eğrilerinin I. bölgede sınırladığı bölge ise

$y=x+3$

$$\iint_D (x^2-y^2)\,dy\,dx = ?$$

$x-y=u$

$xy=v$

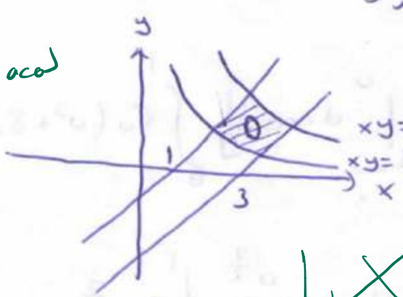
$(x-y)(x+y)=x^2-y^2$

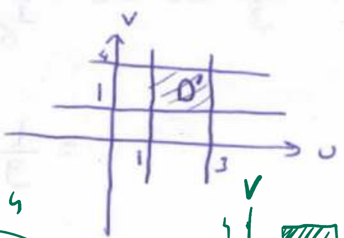
$\frac{1}{x+y} = \text{Jacob}$

$y=\frac{1}{x}$

$\left.\begin{array}{l} x-y=u \\ xy=v \end{array}\right\} \Rightarrow D': \begin{array}{ll} u=1 & u=3 \\ v=1 & v=4 \end{array}$

$(x-y)^2 = x^2-2xy+y^2$





$$\int_1^4\int_1^3 u.\sqrt{u^2+4v}$$

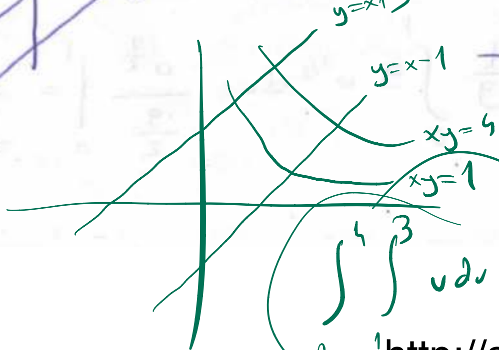
$u^2 = -2v$

$u^2+2v = x^2+y^2$

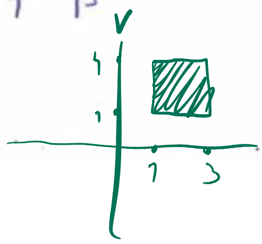
$u\,(x+y)^2 = 2xy + u^2 + 2v$

$\sqrt{u^2+4v}$

$$\begin{vmatrix} 1 & -1 \\ y & x \end{vmatrix}$$



$$\int_1^4\int_1^3 u\,du\,dv$$



i.K.12

$$J=\frac{\partial(x,y)}{\partial(u,v)}=\frac{1}{\frac{\partial(u,v)}{\partial(x,y)}}=\frac{1}{\begin{vmatrix}\frac{\partial u}{\partial x} & \frac{\partial u}{\partial y}\\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y}\end{vmatrix}}=\frac{1}{\begin{vmatrix}1 & -1\\ y & x\end{vmatrix}}=\frac{1}{x+y}$$

$$\iint_D (x^2-y^2)\,dx\,dy=\iint_{D'}(x^2-y^2)\cdot\frac{1}{x+y}\,du\,dv$$

$$=\iint_{D'}(x-y)\,du\,dv=\int_1^3\int_1^4 u\,dv\,du=12$$

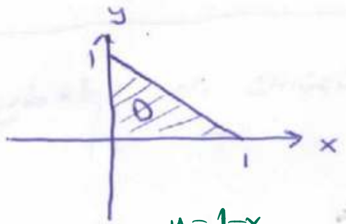
(*) $\displaystyle\int_0^1\int_0^{1-x}\sqrt{x+y}\cdot(y-2x)^2\,dy\,dx=?$ $\qquad \left.\begin{array}{l}u=x+y\\ v=y-2x\end{array}\right\}$ dönüşümü yapalım.

$$\Downarrow$$

$x=\dfrac{u-v}{3}\qquad y=\dfrac{2u}{3}+\dfrac{v}{3}$

$$\begin{vmatrix}1 & 1\\ -2 & 1\end{vmatrix}=3 \qquad \frac{1}{3}=J$$

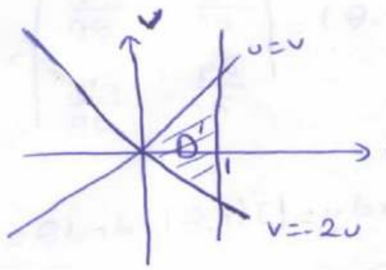
$D:\left.\begin{array}{l}x=1\\ x=0\\ y=1-x\\ y=0\end{array}\right\}$



$y=1-x$

$x+y=1 \qquad u=1$

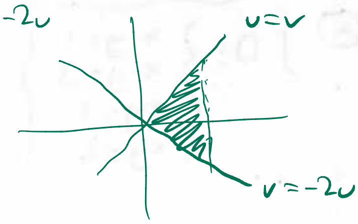
$x=0 \qquad u=v$



D		D'		
$x+y=1$	$\rightarrow$	$\boxed{u=1}$		
$x=0$	$\rightarrow$	$u-v=0$	$\rightarrow$	$\boxed{u=v}$ $y=0$
$y=0$	$\rightarrow$	$2u+v=0$	$\rightarrow$	$\boxed{v=-2u}$ $u=x$, $v=-2x$

$-2u=v$

$$J(u,v)=\begin{vmatrix}\frac{\partial x}{\partial u} & \frac{\partial x}{\partial v}\\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v}\end{vmatrix}=\begin{vmatrix}\frac{1}{3} & -\frac{1}{3}\\ \frac{2}{3} & \frac{1}{3}\end{vmatrix}=\frac{1}{9}+\frac{2}{9}=\frac{1}{3}$$



$$I=\int_0^1\int_{-2u}^{u}\sqrt{u}\cdot v^2\cdot\frac{1}{3}\,dv\,du=\frac{1}{3}\int_0^1\sqrt{u}\cdot\frac{v^3}{3}\Big|_{-2u}^{u}\,du=\frac{1}{9}\int_0^1\sqrt{u}\,(u^3+8u^3)\,du$$

$$\int_0^1\int_{-2u}^{v}\sqrt{v}\cdot v^2\frac{1}{3}\,dv\,du \qquad =\frac{1}{9}\int_0^1 9\,u^{7/2}\,du=\frac{u^{\frac{9}{2}}}{\frac{9}{2}}\Big|_0^1=\frac{2}{9} \qquad \frac{2u^{9/2}}{9}\Big|_0^1$$

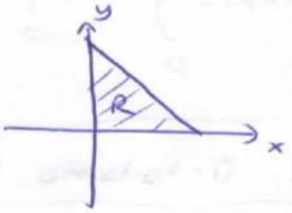
$$\frac{v^3\sqrt{u}}{9}\Big|_{-2u}^{u} \qquad \frac{u^{7/2}}{9}+\frac{8u^{7/2}}{9} \qquad \int_0^1 u^{7/2} \qquad \frac{2}{9}$$

(i.K.13)

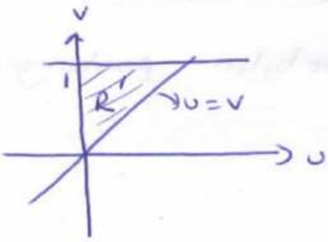
(*) $\iint_R e^{\frac{x}{x+y}}\,dx\,dy = ?$ $\quad R: \left\{ \begin{array}{l} y=0 \\ x=0 \\ x+y=1 \end{array} \right\}$

$\left.\begin{array}{l} x=u \\ x+y=v \end{array}\right\}$ dönüşümü yapalım.

$R: \left\{ \begin{array}{l} x=0 \\ y=0 \\ x+y=1 \end{array} \right\} \Rightarrow$



R		R'
$x=0$	$\to$	$u=0$
$y=0$	$\to$	$u=v$
$x+y=1$	$\to$	$v=1$



$$J(u,v)=\frac{1}{J(x,y)}=\frac{1}{\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}}=\frac{1}{\begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix}}=1$$

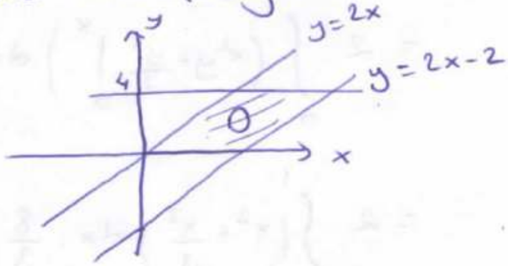
$$\iint_R e^{\frac{x}{x+y}}\,dx\,dy=\iint_{R'} e^{u/v}\,du\,dv=\int_0^1\int_0^v e^{u/v}\,du\,dv$$

$$=\int_0^1 \frac{e^{u/v}}{\frac{1}{v}}\Big|_0^v\,dv=\int_0^1 v(e-1)\,dv=\frac{v^2}{2}(e-1)\Big|_0^1=\frac{e-1}{2}$$

(*) $\displaystyle\int_0^4\int_{y/2}^{(y/2)+1} \frac{2x-y}{2}\,dx\,dy$ integralini $u=\frac{2x-y}{2}$, $v=\frac{y}{2}$ dönüşümü ile hesaplayın.

$y=4 \quad x=\frac{y}{2}+1$

$y=0 \quad x=\frac{y}{2}$



$u=\frac{2x-y}{2}$

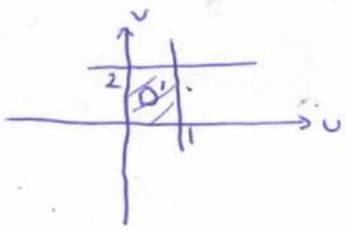
$v=\frac{y}{2}$

$\Downarrow$

$u+v=x$

$2v=y$

D		D'
$y=4$	$\longrightarrow$	$v=2$
$y=0$	$\longrightarrow$	$v=0$
$y=2x$	$\longrightarrow$	$u=0$
$y=2x-2$	$\longrightarrow$	$u=1$



İ.K.14

$$\left.\begin{array}{l} x=u+v \\ y=2v \end{array}\right\} \Rightarrow J(u,v)=\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 2 \end{vmatrix} = 2$$

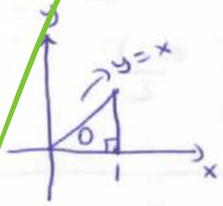
$$I=\iint_{D'} 2.u\,du\,dv=\int_0^1\int_0^2 2u\,dv\,du=\int_0^1 2uv\Big|_0^2 du=\int_0^1 4u\,du=2u^2\Big|_0^1=\underline{\underline{2}}$$

İki Katlı İntegraller İçin Ortalama Değer Teoremi

Bir D bölgesi üzerinde $f(x,y)$ integrallenebilir fonksiyonunun ortalama değeri:

$$\bar{f}=\frac{1}{D\text{'nin Alanı}}\cdot\iint_D f(x,y)\,dA \quad \text{dır.}$$

⊛ Köşeleri $(0,0),(1,0)$ ve $(1,1)$ de olan dik üçgende x^2+y^2 fonksiyonunun Ortalama değerini bulunuz.



$$\bar{f}=\frac{1}{D\text{'nin Alanı}}\cdot\iint_D (x^2+y^2)\,dy\,dx$$

$$=\frac{1}{\frac{1}{2}}\cdot\int_0^1\int_0^x (x^2+y^2)\,dy\,dx$$

$$=2\int_0^1\left(x^2y+\frac{y^3}{3}\Big|_0^x\right)dx$$

$$=2\int_0^1\left(x^3+\frac{x^3}{3}\right)dx=\frac{8}{3}\cdot\frac{x^4}{4}\Big|_0^1=\underline{\underline{\frac{2}{3}}}$$

$$\frac{\int_0^1\int_0^x x^2+y^2\,d_y d_x}{\int_0^1\int_0^x d_y d_x}$$

$$\frac{x^2}{2}=\frac{1}{2}$$

$$\frac{2}{3}$$